

42.510 Statistical Learning for Data Science

Bagging and Random Forests

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Week 11, Lecture 1

Outline

- Bootstrapping
- Bagging
- Random Forests

Bootstrapping

- The **Bootstrap** or **bootstrapping** is a resampling method used to quantify the uncertainty associated with a statistical learning method (or a given estimator).
- The other resampling method introduced in this course is **cross-validation**.
- The term originates from the expression “*to pull oneself up by one’s bootstraps*”.

Bootstrapping

Example

- We want to invest a given amount of money in two financial assets that yield returns of X and Y , respectively, where X and Y are random quantities.
- We will invest a fraction α of our money in X , and will invest the remaining $1 - \alpha$ in Y .
- We wish to choose the value of α that minimizes the total risk, measured with the variance of our investment, $\text{Var}(\alpha X + (1 - \alpha)Y)$.

Bootstrapping

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- The value of α that minimizes the risk is

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Bootstrapping

Example

- The values of σ_X^2 , σ_Y^2 , and σ_{XY} are unknown.
- We can calculate estimates ($\hat{\sigma}_X^2$, $\hat{\sigma}_Y^2$, and $\hat{\sigma}_{XY}$) using a dataset with observations of X and Y .
- And then estimate $\hat{\alpha}$ as

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}.$$

Bootstrapping

Example

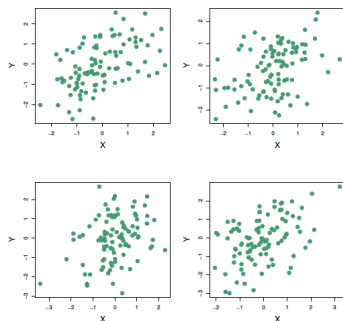


Figure 1: A few simulated returns for X, Y . (Source: James et al., 2023).

- We estimate the necessary values of $\hat{\sigma}_X^2, \hat{\sigma}_Y^2, \hat{\sigma}_{XY}$ from the data.
- The estimates of α from the data are 0.576, 0.532, 0.657, 0.651.

Bootstrapping

Example

We repeat this process 1,000 times, so as to get a better estimate of α :

- We set $\sigma_X^2 = 1$, $\sigma_Y^2 = 1.25$, and $\sigma_{X,Y} = 0.50$, so we know that the true value of α is 0.60.
- The mean over all 1,000 estimates for $\hat{\alpha}$ is

$$\bar{\alpha} = \frac{1}{1000} \sum_{r=1}^{1000} \hat{\alpha}_r = 0.5996.$$

which is close to 0.60. The standard deviation of the estimates is:

$$\sqrt{\frac{1}{1000 - 1} \sum_{r=1}^{1000} (\hat{\alpha}_r - \bar{\alpha})^2} = 0.083.$$

Bootstrapping

Back to the real world ...

- The example above cannot be used, because we cannot generate new samples from the original population.
- **Bootstrapping** mimics this process: we obtain distinct datasets by repeatedly sampling observations from the original data set *with replacement*.
- Each of these bootstrap datasets is created by sampling with replacement, and is the same size as our original dataset.

Bootstrapping

Example (with three observations)

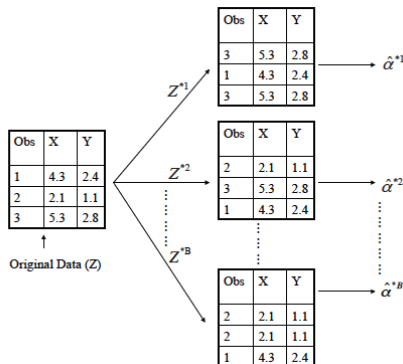


Figure 2: A graphical illustration of the bootstrap approach on a small sample containing $n = 3$ observations. (Source: James et al., 2023).

Bootstrapping

A few more points

- Denoting the first bootstrap data set by Z^{*1} , we use Z^{*1} to produce a new bootstrap estimate for α which we call $\hat{\alpha}^{*1}$.
- The procedure is repeated B times (for example, 100 or 1,000), in order to produce B different bootstrap datasets ($Z^{*1}, Z^{*2}, \dots, Z^{*B}$), and B corresponding estimates $\hat{\alpha}^{*1}, \hat{\alpha}^{*2}, \dots, \hat{\alpha}^{*B}$.
- We estimate the standard error of these bootstrap estimates with the formula

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1} \sum_{r=1}^B (\hat{\alpha}^{*r} - \bar{\hat{\alpha}}^*)^2}.$$

Bootstrapping

What is the difference between cross-validation and Bootstrapping?

- 1 Bootstrap resamples with replacement, while cross-validation resamples without replacement.
- 2 The main goal of cross-validation is to measure, or generalize, the performance of a model.
- 3 Bootstrapping is used to estimate *sampling distribution* for a widespread range of statistics.

Bootstrapping

So, where can we use Bootstrapping in the context of CART?

It can be used to tackle the bias-variance trade-off of some statistical learning methods, such as ... **Decision Trees**.

Bagging

- **Intuition:** Decision Trees suffer from **high variance**
 - ▶ if we split the training data into two parts at random, and fit a decision tree to both halves, the results that we get could be quite different.
 - ▶ **Bagging** (or **Bootstrap aggregation**) can be used to reduce the variance of a statistical learning method.

Bagging

How does it work?

- Given a set of n independent observations $Z_1, Z_2, \dots, Z_n$, each with variance σ^2 , the variance of the mean $\bar{Z}$ of observations is $\frac{\sigma^2}{n}$.
 - ▶ *averaging a set of observations reduces variance.*
- When learning a statistical model, we can therefore:
 - 1 Take many training sets from the population.
 - 2 Build a separate model using each training dataset.
 - 3 Average the resulting predictions.

Bagging

(More formal) steps for the algorithm:

- 1 Take repeated samples from the training dataset (if that's everything we have); we generate B different bootstrapped training datasets.
- 2 We train model $f^{*b}(x)$ on the b -th bootstrapped training dataset (repeat for all B datasets).
- 3 We average all the predictions as follows

$$f_{bag}(x) = \frac{1}{B} \sum_{b=1}^B f^{*b}(x).$$

This entire process is called **Bagging**.

Bagging

How does it work for Classification Trees?

- We only change Step 3:
 - ▶ We record the class predicted by each of the B trees and take a **majority vote**.

Bootstrapping

Can we use Bagging (bootstrapping) to estimate the prediction error?

- We could think about using each bootstrap dataset as our training sample, and the original sample as our validation sample.
- But each bootstrap sample has significant overlap with the original data; hence the bootstrap will underestimate the true prediction error.
- We can fix the problem by using the (*OOB*) *out-of-bootstrap estimate* . . .

Bagging

Out-of-Bag Error Estimation

- To test the error of a bagged model, we can use the out-of-bag (OOB) observations.
- More specifically:
 - ▶ To obtain a single prediction for the i -th observation, we can average these predicted responses (for regression) or can take a majority vote (for classification). This leads to a single OOB prediction for the i -th observation.
 - ▶ An OOB prediction can be obtained in this way for each of the n observations, from which we compute the overall OOB MSE or classification error .
- When B is sufficiently large, we have a decent alternative to cross-validation.

R implementation

How can we implement Bagging in R?

Options:

- 1 Write our own code.
- 2 Use the function `bagging` (package `ipred`). This only works for Decision Trees (and some other algorithms, but not every algorithm).

Random Forests

Problems with using CART (vanilla)

- 1 CART suffers from high variance and low bias.
- 2 When building Decision Trees with Bagging, trees tend to be *correlated*.

Example:

- Suppose the training dataset has a strong predictor and a few moderately strong predictors.
- Then, most of the bagged trees will use the strong predictor in the top split(s).
 - ▶ trees will be correlated, so we won't reduce variance much.

Random Forests

Idea: When building the trees, we consider **a random sample of m predictors** from the full set of predictors p (at each split):

- 1 For each split, we consider only a subset of predictors.
- 2 On average, $(p - m)$ predictors are not considered, so other predictors will have a chance.
- 3 This process decorrelates the trees.

Random Forests

Algorithm

- 1 Generate B bootstrapped training datasets.
- 2 For each dataset, train a Decision Tree. At each split, use a subset m of the p available predictors.
- 3 Average the predictions from the B trees (For classification, use majority voting).

On the value of m

The fundamental difference between Bagging and Random Forests stands in the subset of predictors m :

- If $m = p$, then there is no difference between the two methods.
- Recommended values of m :
 - ▶ Regression: $m = p/3$.
 - ▶ Classification: $m = \sqrt{p}$.
- Note that these values were found experimentally, so there is no theoretical guarantee they will provide the best performance on all datasets.

Random Forests

- **Hyperparameter tuning:** It is common practice to explore the effect of the hyperparameters value on the performance of Random Forests. To recap, we have the following parameters:
 - ▶ Number of trees, B .
 - ▶ Number of predictors used at each split, $m \leq p$.
 - ▶ Number of points in each terminal leaf.
- There are no optimization routines to find their values. We typically use *grid search*, or similar methods.

Back to R!

- To learn a Random Forest, we will use the function `randomForest`, implemented in the package `randomForest`:
 - ▶ `randomForest(formula, data, ntree, mtry, ...)`

Advantages and Disadvantages of Random Forests

Pros:

- Better bias-variance trade-off than CARTs.
- Higher accuracy.

Cons:

- Less interpretable.
- Higher computational requirements.

References

- James et al. (2023) *An Introduction to Statistical Learning with Applications in R*, Chapters 5.2 and 8.2.